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# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
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# Step 2: Load Dataset
# Sample dataset
data = pd.DataFrame({
    'Hours': [1, 2, 3, 4, 5, 6],
    'Scores': [10, 20, 30, 40, 50, 60]
})
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# Step 3: Split features and target
X = data[['Hours']] # input feature
y = data['Scores'] # target variable
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# Step 4: Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) # Split data into training and testing sets
#Added this line to split the data into training and testing sets using train_test_split.
# test_size=0.2 specifies 20% of the data for testing, and random_state=42 ensures consistent results.
# Step 5: Create and train the model
model = LinearRegression()
model.fit(X_train, y_train)
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↗ LinearRegression ⓘ ?
LinearRegression()
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# Step 6: Prediction
y_pred = model.predict(X_test)
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# Step 7: Evaluation
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error:", mse)
print("Slope (Coefficient):", model.coef_[0])
print("Intercept:", model.intercept_)
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↗ Mean Squared Error: 0.0
Slope (Coefficient): 10.0
Intercept: 0.0
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# Step 8: Visualization
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.xlabel("Hours Studied")
plt.ylabel("Score")
plt.title("Linear Regression - Hours vs Scores")
plt.legend()
plt.grid(True)
plt.show()
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